Block practice · Strings, validation checks and readable code

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Make code easy to read

Design validation checks for a code format

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Practice workbook · Make code easy to read

Giving techniques that improve readability and reasons for using comments.

3 question parts: 1 worked-example retry and 2 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T2-07

1. Question 1 · part a · 1 mark

2021 NOV · Paper 02

Worked-example retry · Attempt part a only.

1 Programmers write code to solve problems.

(a) Give **one** technique that programmers use to make their code easy to read and understand.

(1)

.....

(b) Programs may contain logic, syntax or runtime errors.

A program has three errors.

These may be logic, syntax or runtime errors.

Complete the table by adding a tick (✓) for **each** error description to show whether the error is a logic, syntax or runtime error.

(3)

Error description	Logic	Syntax	Runtime
Divide by 0			
Use x instead of * to multiply			
Subtract 10 from 2 instead of 2 from 10			

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1. Question 1 · part a · 1 mark · Working and answer

Working and answer

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2. Question 1 · part d · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part d only.

1 Computer programs use several different programming constructs.

(a) Identify the **most appropriate** data type for the value 4.5

(1)

- ☒ **A** Boolean
- ☒ **B** Character
- ☒ **C** Real
- ☒ **D** String

(b) Give a definition of a logic error.

(1)

(c) Open **Q01c** in the code editor.

The program should print out a counter and the counter with 7 added to it.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01cFINISHED** with the correct file extension for the programming language.

(3)

(d) Give **one** reason why computer programmers should use comments in their programs.

(1)

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2

P 6 1 8 8 1 R A 0 2 1 6

2. Question 1 · part d · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part d only.

(e) Describe the difference between a local variable and a global variable in a computer program.

(2)

(f) Trevor is the manager of a shoe shop.

Sales assistants can earn a bonus based on the numbers of pairs of shoes they sell and the total income for the shop each day.

Open **Q01f** in the code editor.

Amend the code to complete the 'if statement' used to produce the outputs described in the table.

Condition	Output
Shop income is more than £5000 or sales assistant has sold at least 10 pairs of shoes	Bonus is 10% of salary
Shop income is £2000 or more and sales assistant has sold at least 5 pairs of shoes	Bonus is 5% of salary

Do not add any further functionality.

Save your amended code as **Q01fFINISHED** with the correct file extension for the programming language.

(4)

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2. Question 1 · part d · 1 mark · Working and answer

Working and answer

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3. Question 1 · part b · 1 mark

2022 JUN · Paper 02

Further practice · Attempt part b only.

1 Programmers write code to solve problems.

(a) Identify the description of a variable in a computer program.

(1)

- ☒ **A** A value that cannot be used more than once
- ☒ **B** A value that must be input
- ☒ **C** A value that is always used in a calculation
- ☒ **D** A value that can change

(b) Identify the technique that improves the readability of code.

(1)

- ☒ **A** Using indents on every line
- ☒ **B** Using descriptive names for variables
- ☒ **C** Using the correct operators
- ☒ **D** Using suitable data structures

(c) Complete the table by giving an example value for **each** data type.

The first row has been completed for you.

(2)

Data type	Example value
integer	12
char	
real	

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3. Question 1 · part b · 1 mark · Working and answer

Working and answer

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Practice workbook · Design validation checks for a code format

What validation is, and devising validation tests for a structured code/ID format with erroneous data that fails only that test.

3 question parts: 1 worked-example retry and 2 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T2-10

1. Question 5 · part a · 4 marks

2020 NOV · Paper 02

Worked-example retry · Attempt part a only.

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5 Cerys Jones is the manager of a shop that sells electrical goods.

She is creating a stock control program.

(a) Each item of stock will have a ten-character code.

Each code consists of three parts.

Part	Description
Characters 1 to 3	Uppercase letters
Characters 4 to 9	Numbers between 1 and 9 inclusive
Character 10	'e' or 'o' indicating whether the sum of the numbers is even or odd

WAS123456o, FDG232324e and VAC129222e are valid codes.

All codes need to be validated.

Complete the table to show **two** additional validation tests.

For each test, give **one** example of erroneous data. The example that you give should fail **only** that test.

(4)

Validation test	Erroneous data
Is the last character 'o' or 'e'?	ABC654321v

(b) Cerys will need an algorithm to search the table of stock data.

(i) State **two** advantages of using a linear search algorithm rather than a binary search algorithm.

(2)

1 .....

.....

2 .....

.....

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1. Question 5 · part a · 4 marks · Working and answer

Working and answer

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2. Question 1 · part g · 4 marks

2019 JUN · Paper 02

Further practice · Attempt part g only.

1 Computer programs use several different programming constructs.

(a) Identify the **most appropriate** data type for the value 4.5

(1)

- ☒ **A** Boolean
- ☒ **B** Character
- ☒ **C** Real
- ☒ **D** String

(b) Give a definition of a logic error.

(1)

(c) Open **Q01c** in the code editor.

The program should print out a counter and the counter with 7 added to it.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01cFINISHED** with the correct file extension for the programming language.

(3)

(d) Give **one** reason why computer programmers should use comments in their programs.

(1)

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2. Question 1 · part g · 4 marks

2019 JUN · Paper 02

Further practice · Attempt part g only.

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(e) Describe the difference between a local variable and a global variable in a computer program.

(2)

(f) Trevor is the manager of a shoe shop.

Sales assistants can earn a bonus based on the numbers of pairs of shoes they sell and the total income for the shop each day.

Open **Q01f** in the code editor.

Amend the code to complete the 'if statement' used to produce the outputs described in the table.

Condition	Output
Shop income is more than £5000 or sales assistant has sold at least 10 pairs of shoes	Bonus is 10% of salary
Shop income is £2000 or more and sales assistant has sold at least 5 pairs of shoes	Bonus is 5% of salary

Do not add any further functionality.

Save your amended code as **Q01fFINISHED** with the correct file extension for the programming language.

(4)

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2. Question 1 · part g · 4 marks

2019 JUN · Paper 02

Further practice · Attempt part g only.

(g) Car registration codes consist of seven characters.

The characters are two upper case letters followed by two numbers and then three upper case letters.

YH62KLV, EY09VPB and CK53PRJ are valid codes.

All codes need to be validated.

Complete the table to show **two** additional validation tests.

For each test give **one** example of erroneous data. The example that you give should fail **only** that test.

(4)

Validation test	Erroneous data
Is it 7 characters?	AB12CD

(Total for Question 1 = 16 marks)

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2. Question 1 · part g · 4 marks · Working and answer

Working and answer

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3. Question 2 · part a · 1 mark

2025 JUN · Paper 02

Further practice · Attempt part a only.

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2 Programs can process numerical data.

(a) Identify the method used to check input data matches set rules.

(1)

- ☒ **A** Abstraction
- ☒ **B** Encryption
- ☒ **C** Iteration
- ☒ **D** Validation

(b) **Figure 2** shows a section from an employee payslip produced by a computer program.

Pay Rate	Hours		Total Pay
£ 18.75	19		£ 356.25
	Tax	20%	£ 71.25
		Income	£ 285.00

Figure 2

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(i) Complete the table to give **one** data type shown in **Figure 2** and give an example value.

(2)

Data type	
Example	

(ii) Complete the table to give an input, a process and an output used in the program to produce the payslip.

(3)

Input	
Process	
Output	

3. Question 2 · part a · 1 mark · Working and answer

Working and answer

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